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# Microstructure and mechanical properties of CoCrFeNiTiAl$_x$ high-entropy alloys

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ABSTRACT

CoCrFeNiTiAl$_x$ ($x$ values in molar ratio, $x$ = 0, 0.5, 1.0, 1.5 and 2.0) high-entropy alloys were prepared using a vacuum arc melting method. The effects of Al addition on microstructure and mechanical properties were investigated. The results show that only a face-centered cubic (FCC) crystal structure phase is observed in the CoCrFeNiTi alloy. The phase composition transforms to stabilized body-centered cubic (BCC) structure phases and typically cast dendrite structure appears when Al is added. The dendrite region is rich in Co, Ni, Ti and Al elements while the interdendrite region is rich in Fe and Cr elements. Subgrains and nanosized precipitates are observed in the as-cast CoCrFeNiTiAl alloy. These CoCrFeNiTiAl$_x$ high-entropy alloys exhibit excellent room-temperature mechanical properties. For CoCrFeNiTiAl$_{1.0}$ alloy, the compressive strength and elastic modulus reach as high as 2.28 GPa and 147.6 GPa, respectively. High density of dimple-like structure is observed from the fracture surfaces of the Al$_0$ alloy, while alloys with Al addition show typical cleavage fractures with river-like patterns and cleavage steps.



## 1. Introduction

Traditional strategy for developing alloys is to select one or two elements as principal components for primary properties and other minor elements incorporated for definite microstructure and properties, such as steels, TiAl intermetallics, bulk metallic glasses (BMG), etc. [1]. Formation of many intermatallic compounds and other complex microstructures may be anticipated using multiple principal elements, resulting in sample brittleness, difficulty in processing and complication in analysis [2,3]. Recently, this paradigm has been broken by high-entropy alloy developed by Yeh et al. [4]. High-entropy alloys are defined as alloys that contain at least five principal elements with each elemental concentration between 5 and 35 at.%. The high-entropy alloys usually form simple solid solution structures rather than many complex phases. With proper composition design, high-entropy alloys can possess multiple excellent properties such as high strength, good ductility and good resistances to wear, oxidation and corrosion, etc. [5–8].

Elemental composition is a critical factor in alloy design, which will definitely affect the alloying behaviors and properties of high-entropy alloys. Many high-entropy alloy systems have been reported recently, and most of them contain Al [4,6–12]. The addition of Al was identified to having considerable effects on the structures and properties of high-entropy alloys due to its large atomic radius. Tong et al. [9,10] have studied Al$_x$CoCrCuFeNi high-entropy alloys and found that the Al contents had obvious influence on phase transformation and mechanical properties. With minor Al addition, the Al$_x$CoCrCuFeNi alloys showed single FCC structure. As the Al addition reached $x$ = 0.8, mixed FCC and BCC eutectic phases appeared. A single ordered BCC structure was obtained for alloys with Al addition higher than $x$ = 2.8. The hardness of the alloys increased gradually from HV 133 to 655 with the Al contents increasing from $x$ = 0 to 3.0. Zhou et al. [11] have researched the effects of Al addition on the microstructures and compressive properties of the Al$_x$(TiVCrMnFeCoNiCu)$_{100-x}$ high-entropy alloys. The results showed that the phase number decreased gradually as the Al content increasing and the compressive strength reached 2.43 GPa with Al content of $x$ = 11.1. Recently, Zhou et al. [6] have investigated the room-temperature mechanical properties of AlCoCrFeNiTi$_x$ alloys with different Ti contents. They found that the strength of the alloys were greatly enhanced, and this phenomenon was attributed to the solid solution strengthen effect of large atomic radius of Ti. However, Al also possesses atomic radius as large as Ti. Therefore, it is worthwhile to investigate whether Al addition could have great effects on the structures and mechanical properties of the CoCrFeNiTiAl$_x$ high-entropy alloys. In this paper, CoCrFeNiTiAl$_x$ high-entropy alloys were fabricated and the effects of Al addition on the microstructure and mechanical properties were investigated.

## 2. Experimental procedures

Alloy ingots with nominal composition of CoCrFeNiTiAl$_x$($x$: molar ratio, $x$ = 0, 0.5, 1.0, 1.5, 2.0, denoted by Al$_0$, Al$_{0.5}$, Al$_{1.0}$, Al$_{1.5}$, Al$_{2.0}$ respectively) were fabricated using arc melt casting in a

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Ti-gettered high-purity argon atmosphere with a water-cooled copper mould. Elemental metal sheets (purity >99%) were utilized as the raw materials. The ingots were remelted four times to improve chemical homogeneity. The crystalline structure of the as-cast samples were characterized by Rigaku Ultima III X-ray diffractometer (XRD). The microstructure of alloys were observed using scanning electron microscope (SEM, JSM-5610LV) and transmission electron microscopy (TEM, JEOL JEM-2010). The chemical compositions of different phases were calculated from the results of electron probe micro-analysis (EPMA: JXA-8100). Room-temperature compressive properties of $\Phi 4\,\mathrm{mm} \times 10\,\mathrm{mm}$ cylindrical samples were tested on MTS810 materials testing machine with a strain rate of 1 mm/min. Three compression tests were performed to obtain the average value.

## 3. Results and discussions

### 3.1. Crystal structure evolution

The XRD patterns of the as-cast CoCrFeNiTiAl$_x$ high-entropy alloys are presented in Fig. 1. Only diffraction peaks corresponding to a FCC crystal structure is observed in the Al$_0$ alloy. The lattice constant of this FCC phase is estimated to be 0.3589 nm by linearity extrapolation method. The Al$_{0.5}$ alloy is composed of two BCC structure phases (marked as “1”and “2” in Fig. 1) and a minor Laves phase. The Laves phase has a lattice constant of 0.2966 nm and can be identified as FeTi type. The Al$_{1.0}$ alloy shows similar XRD pattern with the Al$_{0.5}$ alloy, but its reflections shift to the leftwards slightly. The reflection shift can be attributed to the increment of lattice constant caused by the solid solution of Al. As Al has large atomic radius, the solid solution of Al will lead to lattice deformation and lattice expansion, thus the lattice constant increases and the reflection shift occurs [13]. Only two BCC phases can be detected in the XRD patterns of the Al$_{1.5}$ and Al$_{2.0}$ alloys. Compared with the XRD pattern of the Al$_{1.5}$ alloy, the “2” BCC XRD pattern reflections shift to the lower angles slightly in the Al$_{2.0}$ alloy, while the reflections of “1” BCC phase have no change. The lattice constants of the “1” and “2” BCC phases in the Al$_{2.0}$ alloy are estimated to be 0.2926 and 0.289 nm, respectively.

The XRD results show that the Al addition has remarkable influence on the phase composition of the CoCrFeNiTiAl$_x$ high-entropy alloys. The phase composition transforms from single FCC phase to stabilized BCC phases with Al content increasing from $x = 0$ to 2.0. The structure change of the high-entropy alloy with Al addition can be explained by using the atomic packing efficiency of the FCC and the BCC [14]. As FCC structure has higher atomic packing efficiency

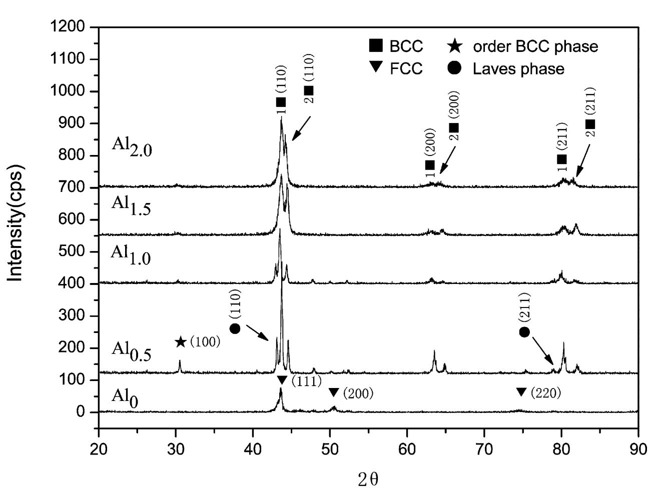


**Fig. 1.** XRD patterns of as-cast CoCrFeNiTiAl$_x$ high-entropy alloys.

(74%) than BCC structure (68%), the solid solution of Al will lead to larger lattice strain and higher lattice distortion energy in FCC structure [15]. To relax the lattice distortion energy, the metastable FCC phase prefers to transform to relatively stabilized BCC structure when Al is added.

### 3.2. Microstructure and chemical composition analysis

Fig. 2 shows the SEM backscattered electron images of the as-cast CoCrFeNiTiAl$_x$ high-entropy alloys. Different morphologies can be observed in the primary phases of these alloys. The Al$_0$ alloy shows morphology of single phase, which is consistent with the XRD result. Alloys with Al addition exhibit a typical cast dendrite structure. The Al$_{0.5}$ alloy shows morphology of primary dendrite phase, chrysanthemum shaped eutectic structure and interdendrite phase, as be marked “Dendrite”, “E(A)” and “Interdendrite” in Fig. 2(b). The Al$_{1.0}$ alloy exhibits the same morphology with Al$_{0.5}$ alloy. This morphology feature indicates that the Al$_{0.5}$ and Al$_{1.0}$ alloys should have similar phase composition and solidification behavior. When Al content reaches $x = 1.5$, the E(A) phase dissolves and only two BCC phases with small dendrite arm width of 5–10 $\mu$m appear. The Al$_{2.0}$ alloy shows the same phase composition with Al$_{1.5}$ alloy, but its cast dendrite morphology changes remarkably. In the Al$_{2.0}$ alloy, large dendrite phases with an average arm width of 20–30 $\mu$m is shown as the main phase while the interdendrite region is composed of long strip shaped minor phase among boundary of the dendrite phases. This phase morphology of the Al$_{2.0}$ alloy is in agreement with the peak intensities in the XRD pattern. The different morphologies shown in Fig. 2 indicate that Al additions have great effects on solidification behavior and microstructure evolution of the CoCrFeNiTiAl$_x$ high-entropy alloys.

Fig. 3 shows the TEM bright-field images and the corresponding select area electron diffraction (SAED) pattern of the Al$_{1.0}$ alloy. The SAED analysis shown in Fig. 3(a) further confirms the BCC structure of the Al$_{1.0}$ alloy. There is an ordered BCC phase in the SEAD pattern, which is in agreement with the XRD pattern. This ordered BCC phase shows a superlattice structure and can be identified as FeTi type. Subgrains and nano-precipitates can be observed in the matrix of the Al$_{1.0}$ alloy in Fig. 3(a). A stack of dislocations tangle along the subgrains and circumambulate around the nano-precipitates. Nanosized ellipsoidal particles (10–50 nm) are clearly identified in the plate structure of the Al$_{1.0}$ alloy in Fig. 3(b). These nanocrystalline structures are rarely seen in the corresponding states of conventional alloys or bulk amorphous alloys [4]. From the viewpoint of kinetics, long-range diffusion for phase separation is sluggish in solid high-entropy alloys with multi-principal elements. Difficulty in substitutional diffusion of elements in these alloys and interactions among interdiffusing species during partitioning lowered the rates of nucleation and growth, leading to the nano-structured crystallites.

The chemical compositions of different phases in the as-cast CoCrFeNiTiAl$_x$ high-entropy alloys are listed in Table 1. Since the Al$_0$ alloy is composed of single FCC phase, it shows a chemical composition similar to the originally designated to the alloy. Elemental segregation can be observed when Al is added. For the Al$_{0.5}$, Al$_{1.0}$, Al$_{1.5}$ and Al$_{2.0}$ alloys, the dendrite region is in rich of Co, Ni, Ti and Al elements while the interdendrite region is in rich of Fe and Cr elements. The E(A) phase shows similar elemental composition to the interdendrite region. The elemental segregation can be explained by the mixing enthalpies among the principal metallic elements [12,16]. From the mixing enthalpy of binary equiatomic alloys listed in Table 2, Cr and Fe have relatively higher mixing enthalpies with other elements. Thus Cr and Fe is repelled from the dendrite region by the majority of the dendrite elements, which makes the interdendrite region rich in Cr and Fe.

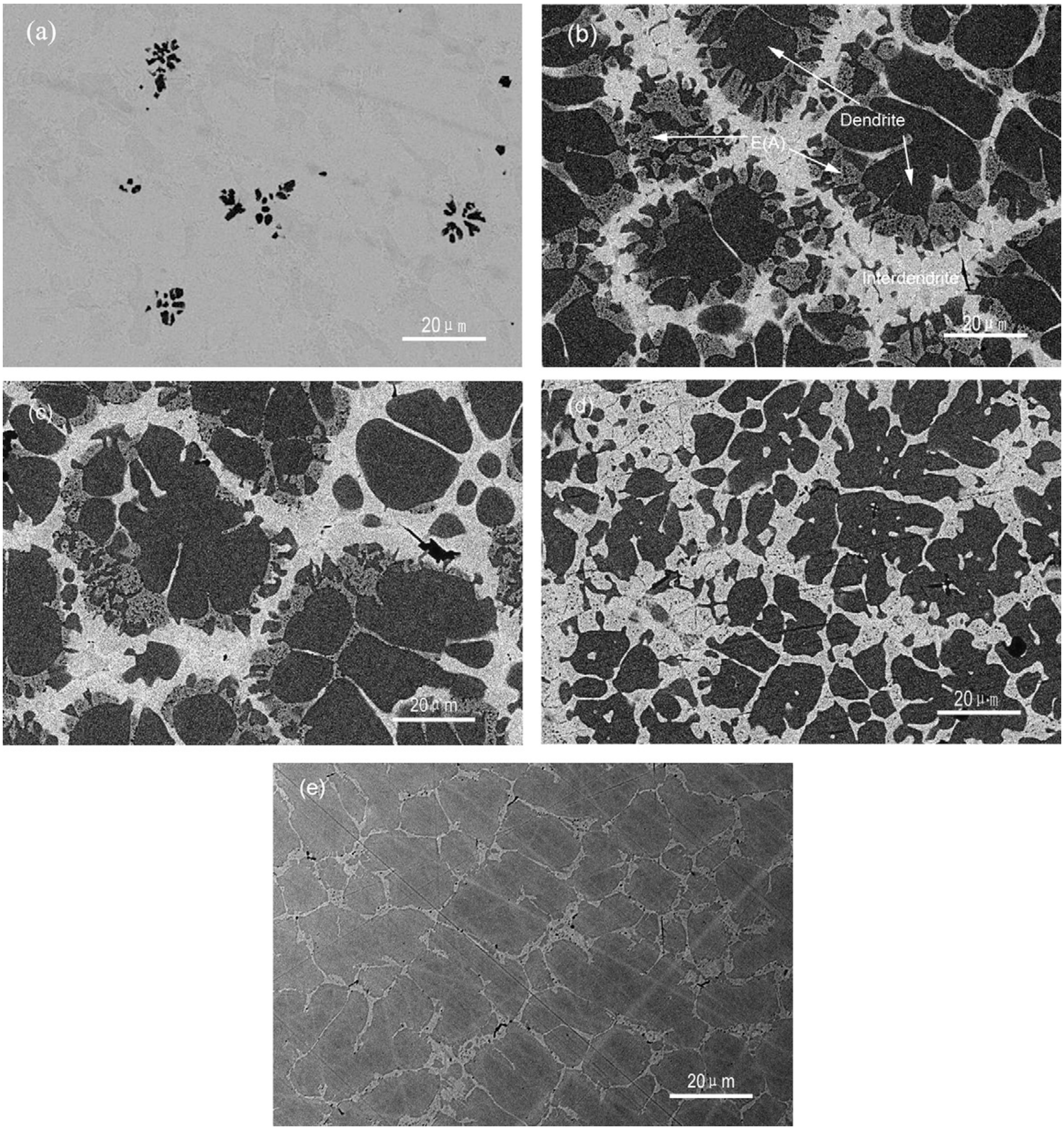


**Fig. 2.** SEM backscattered electron images of the as-cast $CoCrFeNiTiAl_x$ high-entropy alloys: (a) $Al_0$, (b) $Al_{0.5}$, (c) $Al_{1.0}$, (d) $Al_{1.5}$ and (e) $Al_{2.0}$.

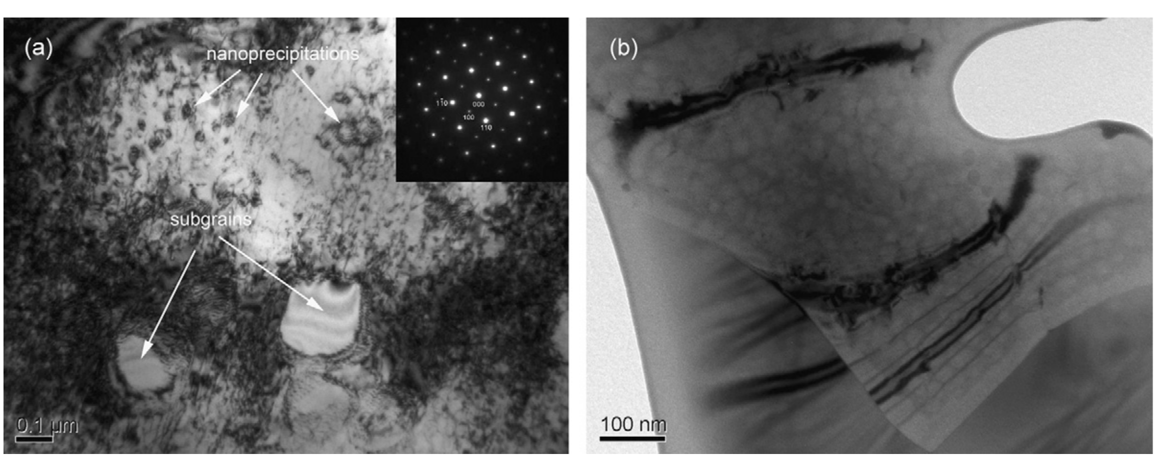


**Fig. 3.** TEM bright-field images and SAED pattern of the $Al_{1.0}$ alloy: (a) bright-field image with SAED pattern of bcc (001) zone axis, (b) bright-field image with nanosized ellipsoidal particles.

**Table 1**
Chemical compositions of as-cast $CoCrFeNiTiAl_x$ high-entropy alloys.

| Alloy | Zone | Co | Cr | Fe | Ni | Ti | Al |
|---|---|---|---|---|---|---|---|
| $CoCrFeNiTiAl_0$ | Nominal | 20 | 20 | 20 | 20 | 20 | 0 |
| | Alloy | 22.51 | 22.66 | 21.90 | 20.97 | 16.39 | 0 |
| $CoCrFeNiTiAl_{0.5}$ | Nominal | 18.2 | 18.2 | 18.2 | 18.2 | 18.2 | 9.0 |
| | E(A) | 13.62 | 36.18 | 26.93 | 10.45 | 8.64 | 3.04 |
| | Dendrite | 24.66 | 5.13 | 9.82 | 26.62 | 20.36 | 12.62 |
| | Interdendrite | 15.55 | 27.58 | 31.63 | 10.78 | 12.75 | 1.97 |
| $CoCrFeNiTiAl_{1.0}$ | Nominal | 16.6 | 16.6 | 16.7 | 16.7 | 16.7 | 16.7 |
| | E(A) | 14.71 | 33.52 | 27.39 | 12.53 | 8.54 | 6.36 |
| | Dendrite | 21.91 | 13.08 | 14.44 | 22.47 | 17.51 | 20.31 |
| | Interdendrite | 16.41 | 24.92 | 29.33 | 14.11 | 13.25 | 5.24 |
| $CoCrFeNiTiAl_{1.5}$ | Nominal | 15.4 | 15.4 | 15.4 | 15.4 | 15.4 | 23.0 |
| | Dendrite | 22.87 | 9.49 | 14.44 | 22.51 | 17.40 | 24.55 |
| | Interdendrite | 15.90 | 23.27 | 22.18 | 15.12 | 14.20 | 13.89 |
| $CoCrFeNiTiAl_{2.0}$ | Nominal | 14.3 | 14.3 | 14.3 | 14.3 | 14.3 | 28.5 |
| | Dendrite | 17.64 | 9.63 | 11.84 | 18.81 | 17.29 | 30.63 |
| | Interdendrite | 8.75 | 42.89 | 28.87 | 4.79 | 5.13 | 11.21 |

**Table 2**
The chemical mixing enthalpy ($\Delta H^{chem}$) of binary equiatomic alloys calculated by Miedema's approach [16].

| Element (atomic size, Å) | Co (1.25) | Cr (1.25) | Fe (1.24) | Ni (1.24) | Ti (1.44) | Al (1.43) |
|---|---|---|---|---|---|---|
| Co | – | −7 | −1 | 0 | −42 | −19 |
| Cr | | | −1 | −7 | −11 | −10 |
| Fe | | | | −2 | −25 | −11 |
| Ni | | | | | −52 | −22 |
| Ti | | | | | | −30 |

### 3.3. Mechanical properties

Fig. 4 shows the room-temperature compressive true stress–strain curves of the as-cast $CoCrFeNiTiAl_x$ high-entropy alloys. The detailed data's of the room-temperature mechanical properties are listed in Table 3. The $Al_0$ alloy exhibits high compressive strength and plasticity strain of 2.02 GPa and 9.0%, respectively, as well as high elastic modulus of 134.6 GPa. The high strength of $Al_0$ alloy can be attributed to the solid solution strengthen of Ti atom and strong bonding effect among the alloy metallic elements [15]. Compared with the $Al_0$ alloy, the $Al_{0.5}$ alloy shows similar ductility while the compressive strength and elastic modulus decrease slightly. This phenomenon can be attributed to the different crystal structure when Al is added. As the $Al_{0.5}$ alloy is mainly composed of BCC structure phases, it has lower atomic packing efficiency and lattice strain than the FCC structure of the $Al_0$ alloy. According to the basic structure factor, a slip in the closest packed FCC structure is more difficult than in the BCC structure. On the other hand, the FCC structure with higher lattice strain will restrain the propagation of dislocations more effectively [17]. Thus the $Al_{0.5}$ alloy with BCC structure exhibits lower compressive strength and elastic modulus than the $Al_0$ alloy. The compressive strength and elastic modulus of $Al_{1.0}$ alloy reach the peak values of 2.28 GPa and 147.6 GPa, respectively. The mechanical property enhancements of $Al_{1.0}$ alloy can be attributed to the increased strengthen effect of lattice strain caused by the lattice sites occupation of Al. The compressive strength and elastic modulus decrease gradually with Al contents increasing from $x = 1.0$ to 2.0. This may results from the saturation of Al solid solution with Al content more than $x = 1.0$. As metal Al is inherently weak in strength, the excessive addition of Al is detrimental to the strength of alloys. Besides high strength, all as-cast $CoCrFeNiTiAl_x$ high-entropy alloys show good ductility as well.

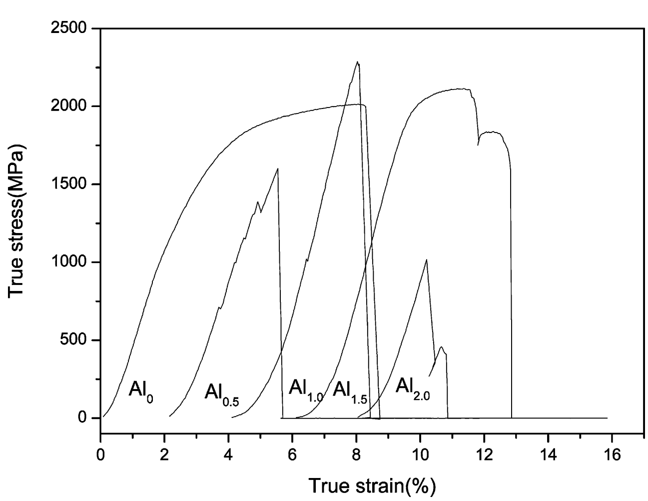


**Fig. 4.** Compressive true stress–strain curves of the as-cast $CoCrFeNiTiAl_x$ high-entropy alloys.

Different fracture surface morphologies can be observed for $CoCrFeNiTiAl_x$ high-entropy alloys (Fig. 5). High density of dimple-like structures are distributed in the fracture surfaces of the $Al_0$ alloy, which would be beneficial for plasticity. However, typical cleavage fracture with river-like patterns and cleavage steps can be observed from the fracture surfaces of the other four alloys. As $Al_0$ alloy is composed of single FCC structure phase, it has different fracture pattern from the other four alloys with BCC structure phases

**Table 3**
Room-temperature compressive properties of as-cast $CoCrFeNiTiAl_x$ high-entropy alloys.

| Alloy | $E$ (GPa) | $\sigma_{max}$ (GPa) | $\varepsilon_p$ (%) |
|---|---|---|---|
| $Al_0$ | 134.6 | 2.02 | 9.0 |
| $Al_{0.5}$ | 106.8 | 1.60 | 9.9 |
| $Al_{1.0}$ | 147.6 | 2.28 | 6.4 |
| $Al_{1.5}$ | 133.4 | 2.11 | 9.8 |
| $Al_{2.0}$ | 93.5 | 1.03 | 5.2 |

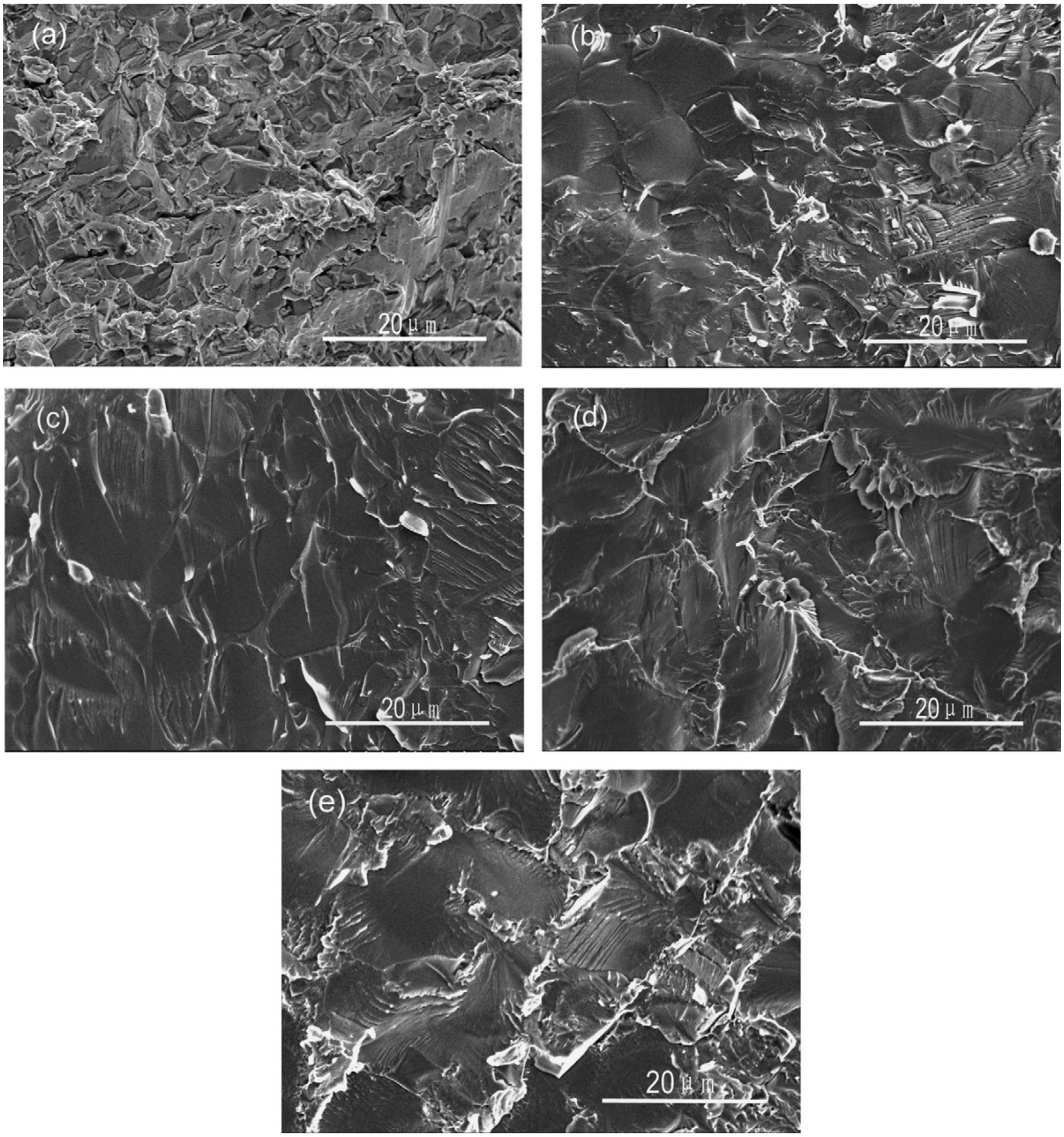


**Fig. 5.** Fracture surfaces of the as-cast CoCrFeNiTiAl$_x$ high-entropy alloys: (a) $Al_0$, (b) $Al_{0.5}$, (c) $Al_{1.0}$, (d) $Al_{1.5}$ and (e) $Al_{2.0}$

[18]. These differences in fracture patterns leads to different fracture morphologies. Both the dimple-like structures and cleavage fracture indicate good ductility of the CoCrFeNiTiAl$_x$ high-entropy alloys.

## 4. Conclusions

CoCrFeNiTiAl$_x$ high-entropy alloys were prepared with simple FCC or BCC crystal structures. The phase composition transforms from single FCC structure phase to stabilized BCC structure phases and typical cast dendrite structure appears when Al is added. Subgrains and nanosized precipitates are observed in the as-cast $Al_{1.0}$ alloy. EPMA results show that the dendrite segregation region is in rich of Co,Ni,Ti and Al elements while the interdendrite segregation region is in rich of Fe and Cr elements. The E(A) phase appeared in $Al_{0.5}$ and $Al_{1.0}$ alloys shows similar elemental composition to the interdendrite region. The compressive strength and elastic modulus of $Al_{1.0}$ alloy reach the peak values of 2.28 GPa and 147.6 GPa, respectively. All as-cast CoCrFeNiTiAl$_x$ high-entropy alloys exhibit good ductility. High density of dimple-like structure is observed from the fracture surfaces of the $Al_0$ alloy, while alloys with Al addition show typical cleavage fracture with river-like patterns and cleavage steps.

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